

B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

SE 19-206-0603 CHEMICAL PROCESS SAFETY
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Differentiate inherent safety and engineered safety and recognize the importance of safety in the design of chemical process plants.
- CO2: Explain the safety aspects of chemical plant operation.
- CO3: Describe the safety precautions to be taken in the storage and handling of gases.
- CO4: State the conditions that lead to reaction hazards and measures to prevent them.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer **ALL** questions)

(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) Discuss the importance of P and ID in a process industry.	3	L2	1	1
(b) Explain the purpose and procedure for the inspection of unfired pressure vessels.	3	L2	1	1
(c) Discuss about NFPA diamond. Discuss the significance of NFPA diamond.	3	L1	2	1
(d) Explain the standard operating procedure of a chemical reactor.	3	L1	2	1
(e) What is sizing of relief valves? What are the data required for this?	3	L1	3	1
(f) Discuss the term 'Implosion' with respect to a storage tank.	3	L2	3	1
(g) What are the mechanisms and hazards of self-heating of solids?	3	L1	4	1
(h) What is the concept of thermochemical screening?	3	L2	4	1

PART B

(4 × 12 = 48)

II. (a) Explain the non-destructive testing methods for pressure vessels.	6	L1	1	1
(b) Discuss briefly about the steps involved in start up procedure of a plant after maintenance.	6	L1	1	1
OR				
III. (a) Write a brief description about safe handling of chemicals in laboratories.	6	L2	2	1
(b) What are the hazard scenarios during the pipeline transportation of flammable gases and liquids?	6	L3	2	1
IV. What is MSDS? Prepare a sample MSDS for methane gas.	12	L3	2	1
OR				
V. (a) Describe various types of relief valves and their operating principles.	6	L2	2	1
(b) Write a note on "Residual life assessment".	6	L3	2	

(P.T.O.)

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		Marks	BL	CO	PO
<u>VI.</u>	Explain various types of storage tanks used in petroleum depots. Discuss the various safety features of these tanks.	12	L1	3	1
OR					
VII.	Write short notes on: (i) Flow indicators (ii) Lightning arresters (iii) Atmospheric venting (iv) Static electricity.	12	L2	3	1
OR					
VIII.	(a) Write a note on colour coding of pipelines.	6	L2	4	1
	(b) Explain hazardous area classification and its significance.	6	L1	4	1
OR					
IX.	(a) Describe briefly about the hazardous organic and inorganic reactions.	6	L1	4	1
	(b) Write short notes on: (i) Thermochemical screening (ii) Self heating hazards of solids.	3 3	L2	4	1

Blooms's Taxonomy Level's

L1 - 40%, L2 - 40%, L3 - 20%.
